Northstar Field Sensor NS-200: User Manual

Synthetic retrieval-test document. Northstar and all figures in it are fictional.

Manual revision DOC-PRD-2026-01. This manual covers the NS-200 field sensor, a solar-assisted device that measures soil moisture, air temperature and humidity and reports readings over a cellular network.

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Installation

Mount the NS-200 on a pole between 1.2 and 1.5 metres above the ground, with the solar panel facing south. Insert the soil probe vertically to a depth of 30 cm. Keep the sensor at least 5 metres away from irrigation sprinklers.

Press and hold the power button for 3 seconds. The device registers with the network within 10 minutes. Pair it in the Northstar Field app by scanning the QR code printed inside the battery cover.

Battery and charging

The NS-200 uses a 6,000 mAh lithium iron phosphate battery. With the default reporting interval of 15 minutes, a full charge lasts 21 days without sunlight. Reducing the interval to 5 minutes shortens battery life to about 8 days.

The device charges from the built-in solar panel, or from a USB-C charger rated at 5 V and 2 A. A full charge over USB-C takes about 4 hours. Do not charge the battery when the air temperature is below 0 degrees Celsius.

Status lights and troubleshooting

Solid green: connected and reporting. Blinking green: searching for the network. Blinking amber: battery below 20 percent. Solid red: sensor fault.

If the light is solid red, remove the soil probe, clean the contacts with a dry cloth and reinsert it. If the fault remains, perform a factory reset by holding the power and reset buttons together for 10 seconds. Error code E-17 means the SIM card is not detected; reseat the SIM card and restart the device.

Warranty and support

The NS-200 has a 24-month limited warranty from the date of delivery. The warranty does not cover physical damage, water entering through an open battery cover, or batteries older than 12 months.

To make a claim, open a support ticket with the category PRD-WAR and include the serial number printed on the back of the device. Replacement units are shipped within 5 working days of an approved claim.